

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2024 INVITATIONAL B**

Biology

B01. D
B02. E
B03. C
B04. D
B05. B
B06. B
B07. C
B08. A
B09. A
B10. C
B11. D
B12. E
B13. A
B14. E
B15. B
B16. D
B17. A
B18. B
B19. E
B20. C

Chemistry

C01. D
C02. E
C03. B
C04. B
C05. C
C06. E
C07. A
C08. B
C09. B
C10. E
C11. E
C12. B
C13. D
C14. C
C15. A
C16. A
C17. E
C18. D
C19. C
C20. E

Physics

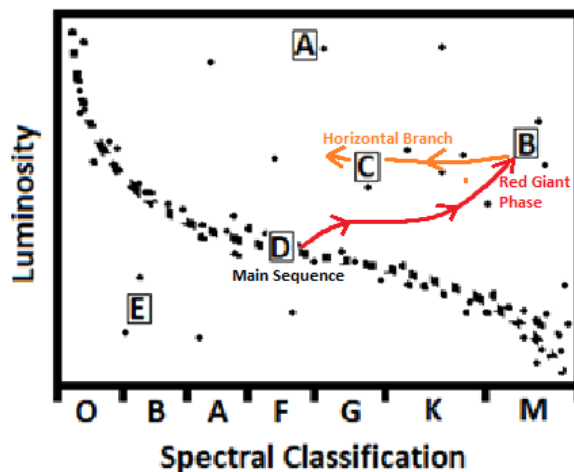
P01. E
P02. D
P03. B
P04. C
P05. C
P06. A
P07. C
P08. D
P09. B
P10. B
P11. D
P12. E
P13. A
P14. A
P15. A
P16. C
P17. E
P18. D
P19. B
P20. D

CHEMISTRY SOLUTIONS – UIL INVITATIONAL B 2024

- C01. (D) 4.00×10^{23} atoms / 6.02×10^{23} atoms/mol = 0.66445 mol $\times$ 4.00 g/mol = 2.66 grams
- C02. (E) One mole of Mg produces one mole of H₂ gas. One mole of Mg is 24.31 grams, and 1 mole of any gas at STP takes up 22.4 L.
- C03. (B) The barium in BaCl₂ is Ba²⁺, so it is missing two electrons from its neutral count of 56, leaving it with 54 electrons.
- C04. (B) Covalent bonds are typically formed between two nonmetals.
- C05. (C) Doubling the temperature in Kelvins would double the pressure, but the temperature change in this case is not doubled in Kelvins, it goes from 323 K to 373 K, an increase of 15.5%. The pressure would therefore increase by 15.5%, from 2.0 atm to 2.31 atm. $P_2/P_1 = T_2/T_1$
- C06. (E) CH₄ is non-polar and has only dispersion forces, compounds B-D all have dipole-dipole forces in addition to dispersion forces, and methanol (CH₃OH) has hydrogen bonding as well.
- C07. (A) The heat absorbed by the water = $q = mc\Delta T = (200)(4.184)(6.5) = 5439$ J, so that much heat was given off by the reaction.
- C08. (B) $20.0 \text{ g} / 58.44 \text{ g/mol} = 0.3422 \text{ mol}$. $0.3422 \text{ mol} / 0.25 \text{ kg H}_2\text{O} = 1.369 \text{ molal}$.
 $\Delta T = ik_b m = 2 \times 0.512 \times 1.369 = 1.40^\circ\text{C}$. $T_{\text{bp}} = 100 + 1.40 = 101.4^\circ\text{C}$
- C09. (B) The chemical reaction is $\text{Zn}^{2+}(aq) + 4 \text{NH}_3(aq) \rightarrow \text{Zn}(\text{NH}_3)_4^{2+}(aq)$, so the equilibrium constant is products over reactants, each raised to their stoichiometric coefficients.
- C10. (E) $\text{pOH} = -\log[\text{OH}^-] = -\log(8.95 \times 10^{-6}) = 5.05$. $\text{pH} = 14 - \text{pOH} = 14 - 5.05 = 8.95$
- C11. (E) $K_{\text{sp}} = [\text{Ni}^{2+}][\text{OH}^-]^2 = [x][2x]^2 = 4x^3$. $x = (2.8 \times 10^{-16}/4)^{1/3} = 4.13 \times 10^{-6}$ $[\text{OH}^-] = 2x = 8.3 \times 10^{-6}$.
- C12. (B) Hydrogen is being oxidized in A and E, is neither oxidized nor reduced in C and D, and is reduced in B.
- C13. (D) Reaction rate is in M/s, so if $M/s = k \times M^2$, the units on k must be $M^{-1} \cdot s^{-1}$
- C14. (C) The percent composition of KBr is: Potassium $(39.10/(39.10 + 79.90) \times 100 = 32.9\%$ potassium and 67.1% bromine, so in a 100 gram sample there would be about 33 grams of K and 67 grams of Br, or just over twice as many grams of bromine as potassium.
- C15. (A) The mnemonic *Raging Martians Invaded Venus Using X-Ray Guns* lists the regions of the electromagnetic spectrum from lowest energy (longest wavelength) to highest energy (shortest wavelength): radio, microwave. Infrared, visible, ultraviolet, x-ray, gamma. Of the answer choices, microwaves are the lowest energy and therefore the longest wavelength.
- C16. (A) $PV = nRT$ so $PV = (g/MM)RT$. Therefore mass density = $g/L = P \times MM/RT = 1 \times 38.00/22.4 = 1.70 \text{ g/L}$. Alternatively you could assume one mole of gas at STP, in which case you would have 38 grams in 22.4 liters, which is the same math and comes out to 1.70 g/L.
- C17. (E)
- C18. (D) Ideal gases have no intermolecular attractions between the molecules and a more polar molecule will have stronger IMF's, which increases the van der Waals a constant and leads to non-ideal behavior.
- C19. (C) The volumes of the solutions do not matter here, only the moles of each reactant. Moles of NaOH = $7.5 \text{ g}/58.44 \text{ g/mol} = 0.1283$ moles. Moles of AgNO₃ = $30.0 \text{ g}/169.88 \text{ g/mol} = 0.1766$ moles. NaOH is limiting, producing 0.1283 moles of AgOH $\times 143.32 \text{ g/mol} = 18.4$ grams.
- C20. (E) Actual yield is measured, not calculated. Theoretical yield is calculated from the moles of the limiting reactant.

PHYSICS SOLUTIONS – UIL INVITATIONAL B 2024

- P01. (E) page 72: “The Bernoulli brothers, too, were unable to make much sense of Leibniz’s paper, despite their dogged efforts to do so.... Undiscouraged, they persevered, until one day, as if by some miracle, Jakob suddenly understood everything.... It all hinged on something called the ‘infinitesimal,’ Jakob explained,”
- P02. (D) pages 79-80: “...like his illustrious friend Leibniz, the elder Bernoulli called it *vis viva*... because it appeared to be something possessed by objects that were to some degree animated.... It was as if an object’s *vis viva* could never be destroyed... they called it the ‘Law of *Vis Viva* Conservation’ (Late in the next century, scientists would call it the ‘Law of Energy Conservation’”
- P03. (B) pages 115-116: “Now, two years after the Wright brothers’ stunning achievement, forty-four-year-old Zhukovsky himself was about to fly straight into the history books. Airplanes were able to fly, he announced in 1905, because of Bernoulli’s fluid-flow equation.”
- P04. (C) A star similar in mass to our Sun will become a red giant after running out of hydrogen in its core. The red giant phase is located near the point B on the HR-diagram. When helium fusion initiates, the changes in the star push it along the horizontal branch, so named because the star evolves along a near-horizontal line on the HR-diagram (as shown). Thus, near the point C on the HR diagram is where you would find the horizontal branch stars.



- P05. (C) Following proper order of operation, the subtraction in the numerator would be completed first. During addition or subtraction, it is the quantity with the least significant place that determines the significant place of the result. In our case, the quantity 13.3 is significant to the tenths place, while 9.55 is significant to the hundredths place. The least significant place of these two is the tenths place, so the result will only be significant to the tenths place. That is $13.3 - 9.55 = 3.7[5]$ is only significant to the tenths place. Thus, the result has two significant digits.
- Now we consider the division. During multiplication or division, it is the quantity with the least number of significant digits that determines the significant digits of the result. In our case, $3.7[5]$ has two significant digits, and 56.1 has three significant digits. Thus, the result (and the final answer) will only have two significant digits: $\frac{3.7[5]}{56.1} = 0.066[8449198] \approx 0.067$.
- P06. (A) This is a free fall situation, even though the toy is travelling upward throughout the problem. We aren’t given the time between throwing the toy and hitting the ceiling, so we should use the kinematic equation that doesn’t involve time: $v_f^2 = v_i^2 + 2a(y_f - y_i) = (9.65)^2 + 2(-9.80)(3.35 - 0)$. This gives $v_f^2 = 27.46 \rightarrow v_f = 5.24\text{m/s}$, which is the speed of the toy when it hits the ceiling.

- P07. (C) We begin with a free-body force diagram for each box. For the dangling box, there are two forces: gravity (m_1g , pointed downward) and tension (T , pointed upward). For the box on the table, there are three forces: gravity (m_2g , pointed downward), the normal force (F_N , pointed upward), and tension (T , pointed to the right). The magnitude of the acceleration for both boxes will be the same, but for the dangling box the acceleration is downward, and for the box on the table the acceleration is to the right. Now we use Newton's second law for each of the boxes. For the dangling box, we sum up the vertical forces: $\sum F_y = T - m_1g = m_1(-a) = -m_1a$. Here the acceleration is negative since the acceleration is directed downward, and we are taking "up" to be positive. For the box on the table, we sum the horizontal forces (of which there is only one). Taking "to the right" to be positive: $\sum F_x = T = m_2a$. Plugging in the values for the masses and g , we get $T - (1.80)(9.80) = -1.80a$, and $T = 2.50a$. This simplifies to $T - 17.64 = -1.80a = 2.50a - 17.64 \rightarrow 17.64 = 4.30a \rightarrow a = 4.10\text{m/s}^2$.
- P08. (D) Since this situation involves a collision, we will use conservation of momentum to solve it. Initially, the block is stationary and only the nail is moving. The initial momentum is $p_i = m_nv_{ni} + m_bv_{bi} \rightarrow p_i = (0.032\text{kg})(80.0\text{m/s}) + (0.215\text{kg})(0.0\text{m/s}) = 2.56\text{kgm/s}$. After the collision, both the nail and the block are moving. Thus, the final momentum is $p_f = m_nv_{nf} + m_bv_{bf} \rightarrow p_f = (0.032\text{kg})(43.0\text{m/s}) + (0.215\text{kg})v_{bf} = 1.376 + 0.215v_{bf}$. By conservation of momentum, the initial and final momentum must be the same. That is $p_f = p_i \rightarrow 1.376 + 0.215v_{bf} = 2.56$. This leads to $0.215v_{bf} = 1.184 \rightarrow v_{bf} = 5.51\text{m/s}$.
- P09. (B) First, let's consider the magnitude of the angular acceleration of the merry-go-round. The initial angular velocity is given, and the final angular velocity is zero. The angular distance through which the merry-go-round travels while stopping is $\Delta\theta = 4.00\text{rev} * 2\pi = 25.13\text{rad}$. Now we use the following angular motion kinematic equation: $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta \rightarrow 0 = (36.0)^2 + 2\alpha(25.13)$. This gives an angular acceleration of $\alpha = -25.8\text{rad/s}^2$. The negative sign confirms that it is slowing down, but only the magnitude, $|\alpha| = 25.8\text{rad/s}^2$, is needed in the next calculation. The total torque acting on the merry-go-round is related to the magnitude of the angular acceleration by $\tau = I\alpha$ where I is the moment of inertia. Thus, $|\tau| = I|\alpha| = (8.60)(25.8) = 222\text{Nm}$.
- P10. (B) The relationship between the force on the cable and the change in length of the cable is given by the equation $\frac{F}{A} = Y\frac{\Delta L}{L_0}$. We know the force ($F = 8600\text{N}$), Young's modulus ($Y = 2.0 \times 10^{11}\text{Pa}$), and the original length of the cable ($L_0 = 450.0\text{m}$). We need the cross-sectional area of the cable. Since the cross section is a circle, we can find the area by $A = \pi r^2$. The diameter is 18.0cm, so the radius of the cable is 9.00cm, giving a cross-sectional area of $A = \pi(0.0900\text{m})^2 = 0.0254\text{m}^2$. Putting it all together, we get $\frac{8600.0\text{N}}{0.0254\text{m}^2} = (2.0 \times 10^{11}\text{Pa})\frac{\Delta L}{450.0\text{m}} \rightarrow 338580 = (4.44 \times 10^8)\Delta L$. This gives a change in length (a stretch) of $\Delta L = 7.6 \times 10^{-4}\text{m} = 0.76\text{mm}$.
- P11. (D) For an adiabatic process, we have a modified form of Boyle's Law: $P_1V_1^\gamma = P_2V_2^\gamma$ where the exponent γ is the heat capacity ratio, a value which is given in the problem. Conveniently, we don't need to change any of the units, we just insert the given values and calculate the final pressure: $(125\text{kPa})(0.355\text{L})^{1.40} = P_2(0.610\text{L})^{1.40} \rightarrow 29.32 = 0.501P_2 \rightarrow P_2 = 58.6\text{kPa}$.
- P12. (E) First, we must combine the individual capacitances into a single equivalent capacitance. We begin by combining the two capacitors that are in parallel; recalling that capacitors in parallel add to get the equivalent capacitance we get: $C_p = C_1 + C_2 = 120\mu\text{F} + 140\mu\text{F} = 260\mu\text{F}$. Now, we combine this equivalent capacitance with the third capacitor. These are in series, and capacitances in series add in reciprocal: $\frac{1}{C_T} = \frac{1}{C_p} + \frac{1}{C_3}$. This gives us: $\frac{1}{C_T} = \frac{1}{260\mu\text{F}} + \frac{1}{200\mu\text{F}} \rightarrow C_T = 113\mu\text{F}$. This is the total equivalent capacitance of the circuit. Using this total capacitance, and the voltage provided by the battery, we can find the total charge associated with the circuit: $Q_T = C_TV_T = (113\mu\text{F})(16.0\text{V}) = 1808\mu\text{C}$. At this point, we work back through the circuit. Capacitances in series will have the same charge as their combined capacitance. In other words, the charge on the series capacitances C_p and C_3 will be $Q_p = Q_3 = Q_T = 1808\mu\text{C}$. From this, we can

determine the voltage across the parallel group of capacitors: $V_p = \frac{Q_p}{C_p} = \frac{1808\mu C}{260\mu F} = 6.95V$. For capacitors in parallel, they will all have the same voltage. That is, $V_1 = V_2 = V_p = 6.95V$. Finally, we can calculate the charge stored on the $140\mu F$ capacitor: $Q_2 = C_2V_2 = (140\mu F)(6.95V) = 974\mu C$.

- P13. (A) In this kind of problem, the electric potential can be calculated using the equation $V = \frac{kQ}{r}$ where Q is the charge, k is the Coulomb constant, and r is the distance from the charge to the point of interest (in this case, the origin). Fortunately, electric potential is a scalar, so we calculate the electric potential due to each charge individually and add them to get the total. The electric potential due to the first charge is $V_1 = \frac{kQ_1}{r_1} = \frac{(8.99 \times 10^9 Vm/C)(25.0 \times 10^{-6} C)}{(0.50m)} = 449,500 V$. Similarly, the electric potential due to the second charge is $V_2 = \frac{kQ_2}{r_2} = \frac{(8.99 \times 10^9 Vm/C)(-12.0 \times 10^{-6} C)}{(0.30m)} = -359,600 V$. This potential is negative since the second charge is negative. To get the total electric potential, we add the potentials from each individual charge: $V = (449,500) + (-359,600) = 89,900 V = 89.9 kV$.
- P14. (A) The radius of the circle traced out by a charged particle in a magnetic field is given by the equation $r = \frac{mv}{QB}$. The velocity, v , of the particle is given, as is the strength of the magnetic field, B . The charge of the particle is the same as that of a proton: $Q = 1.602 \times 10^{-19} C$. And the radius is half of the diameter: $r = \frac{D}{2} = \frac{246}{2} = 123cm = 1.23m$. Putting it all together, we get $(1.23m) = \frac{m(2.40 \times 10^5 m/s)}{(1.602 \times 10^{-19} C)(3.60 \times 10^{-3} T)} \rightarrow 1.23 = (4.161 \times 10^{26})m \rightarrow m = 2.96 \times 10^{-27} kg$. The mass of a proton is $m_p = 1.67 \times 10^{-27} kg$, so the mass of the particle as compared to a proton is the ratio $\frac{m}{m_p} = \frac{2.96 \times 10^{-27} kg}{1.67 \times 10^{-27} kg} = 1.77 \rightarrow m = 1.77m_p$.
- P15. (A) Since the north end of the magnet is pointed downward, the magnetic field passing through the loop is also pointed downward (the magnetic field points away from the North end of a magnet). When the magnet is pulled away, the strength of the field decreases. This will induce a current in the loop. The induced current will oppose the change – in this case it will try to strengthen the field that is decreasing. That is, the induced current will be oriented so as to produce a downward pointed magnetic field. We now use the right-hand rule: Pointing your thumb in the direction of the induced field (downward), the curl of your fingers shows the direction of the induced current. In this case, that curl of the fingers (as seen from above) is clockwise around the loop – thus, the correct answer is A.
- P16. (C) The surface of the water functions as a refracting surface, and the equation for an image formed by a refracting surface is $\frac{n_1}{p} + \frac{n_2}{q} = \frac{n_2 - n_1}{R}$. Since the surface is flat and smooth, it has an effective radius of curvature of infinity, and the object distance is given as $p = 45.0cm$. The object (the quarter) is down in the water, so the first index of refraction is that of the water. The second index of refraction, where the observer is located, is that of the air. Thus, $n_1 = 1.33$ and $n_2 = 1.00$. Putting the values that we know into the equation: $\frac{1.33}{45.0} + \frac{1.00}{q} = \frac{1.00 - 1.33}{\infty}$. This gives $\frac{1.33}{45.0} + \frac{1}{q} = 0 \rightarrow \frac{1}{q} = -0.02956 \rightarrow q = -33.8cm$. The negative sign tells us that the image is located below the surface. Thus, we conclude that the image of the quarter is located 33.8cm below the surface.
- P17. (E) We begin by finding the momentum of the electron. Its velocity is much less than the speed of light, so we can use a classical formula for finding the momentum:
 $p = mv = (9.11 \times 10^{-31} kg)(6.25 \times 10^6 m/s) = 5.69 \times 10^{-24} kgm/s$. Now we use de Broglie's formula for the wavelength of a particle: $\lambda = \frac{h}{p}$, where h is Planck's constant.
 This gives $\lambda = \frac{6.626 \times 10^{-34} Js}{5.69 \times 10^{-24} kgm/s} = 1.16 \times 10^{-10} m = 0.166nm$.

- P18. (D) The parent isotope is Nobellium-266; it has $Z = 102$ protons and $N = 266 - 102 = 164$ neutrons. Each alpha decay subtracts two protons and two neutrons from the total. Each negative beta decay changes a neutron into a proton, and each gamma decay carries away energy but doesn't change the isotope. The decay chain includes five alpha decays, two negative beta decays, and one gamma decay. So, the proton number will become $Z = 102 - 5(2)[\alpha] + 2(1)[\beta] + 0[\gamma] = 94$, which is the element Plutonium, Pu.
Likewise, the neutron number will become $N = 164 - 5(2)[\alpha] - 2(1)[\beta] + 0[\gamma] = 152$. Thus, the mass of the daughter isotope will be $A = Z + N = 94 + 152 = 246$. Therefore, the final daughter isotope in this decay chain will be Plutonium-246, ${}^{246}_{94}\text{Pu}$.
- P19. (B) The maximum height is not a linear function of the initial upward velocity. The relationship between the maximum height and the initial velocity is described by the equation $v^2 = v_i^2 + 2a(y - y_i)$. Here $(y - y_i) = h$ is the height. Also, at the maximum height, the velocity v is zero. Thus, $(0)^2 = v_i^2 + 2ah \rightarrow 2ah = -v_i^2$. We can use a single point on the curve to solve for the acceleration. I chose the point $(32.5\text{m/s}, 80\text{m})$. Using the chosen point, we find the approximate acceleration due to gravity on this planet to be: $2a(80\text{m}) = -(32.5\text{m/s})^2 \rightarrow 160a = -1056 \rightarrow a = -6.6\text{m/s}^2$. The negative means that the acceleration is directed downward (as expected), but we only need to report the magnitude $|a| = 6.6\text{m/s}^2$. Using other points on the curve gives results that range from 6.4m/s^2 to 6.8m/s^2 – all of which are close enough to the correct choice.
- P20. (D) Double slit interference is governed by the equation $d \sin \theta = m\lambda$. The first thing we need to determine is the angle subtended by a single interference fringe. Measuring a single fringe is inaccurate, so we will measure ten fringes, and divide by 10 to get the average spacing for a single fringe. Starting with the fringe above the 4-cm mark on the ruler, and counting ten fringes to the right lands us at about the 8.75-cm location on the ruler. Thus, the ten-fringe distance is $Y = 8.75 - 4 = 4.75\text{cm}$. This gives an average single fringe spacing of $y = 4.75\text{cm}/10 = 0.475\text{cm} = 0.00475\text{m}$. Now we can use the distance from the double slit to the screen to find the tangent of the subtended angle:

$$\tan \theta = \frac{y}{L} = \frac{0.00475\text{m}}{2.40\text{m}} = 0.001979$$
This gives a subtended angle of $\theta = \tan^{-1}(0.001979) = 0.1134^\circ$.
Since we are using the spacing for a single fringe, we know that $m = 1$; and we are given that the wavelength of the laser is $\lambda = 633\text{nm}$. Putting it all together, we get: $d \sin \theta = m\lambda \rightarrow$
 $d \sin(0.1134^\circ) = (1)(633\text{nm}) \rightarrow d = 320000\text{nm} = 0.32\text{mm}$, which is the slit separation.